

Contractile elements in the regulation of macromolecular permeability^{1,2}

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It has been well documented that one response to endogenous mediators of edema such as histamine, prostaglandins (PGs), or bradykinin is an effect on vascular smooth muscle to alter regional blood flow patterns. Many of these same mediators also produce edema by increasing protein leakage from the vasculature to the interstitium (4, 29, 43). In addition, macromolecular leakage in response to histamine or bradykinin appears to emanate only from venules or the venular ends of capillaries and does not appear to be uniform along the length of any one venule (17, 34, 37, 38). Even along the length of a single venule there is a heterogeneous distribution of receptors or effectors for macromolecular leakage because repeated applications of bradykinin produce protein leakage at essentially the same vascular sites as in an initial application (20).

It has thus been proposed by numerous investigators that mediators of edema produce protein leakage by altering the contractile state of one or more of the components of the vascular wall, the smooth muscle cell, the endothelial cell, or the pericyte. Inasmuch as protein leakage occurs in the venules that have little or no smooth muscle, it is improbable that vascular smooth muscle itself is involved. This conclusion has been confirmed by microscopic experiments with different vasodilators in the cremaster muscle of the rat (37) and also by studies of limb perfusion in the dog (5, 21). Thus, although many agents that have produced macromolecular leakage (such

as histamine or bradykinin) are also vasodilators, current evidence strongly mitigates against arteriolar or venular dilation and protein leakage as a cause and effect phenomenon.

In 1969 Majno et al. (33) hypothesized that mediators of edema produce contractions of vascular elements such that gaps of up to 1 μm in diameter open up between endothelial cells, which allows plasma macromolecules to leak into the interstitium. In their studies of rat cremaster muscle microvessels, histamine enhanced the leakage of macromolecules into the interstitium. Administration of histamine also produced deformation of the endothelial nucleus, which the investigators interpreted as evidence of endo-

thelial cell contraction. Joris et al. (25) demonstrated that histamine induces bulging of venular endothelial cells of the mesentery. The time course of this response corresponded to that observed for the leakage of carbon particles. Leakage of carbon was not accompanied by changes in venular diameter. Subsequent mor-

ABSTRACT

The leakage of macromolecules from the vasculature to the interstitium is greatly accentuated by mediators of edema such as histamine and bradykinin. The mechanism for this effect is not well delineated although many agents that affect smooth muscle tone may also affect macromolecular leakage. Leakage occurs primarily from the small venules. The demonstration that mediators of edema produce interendothelial gaps in the venules as well as changes in the shape of the endothelial nuclei has led to the hypothesis that a contraction of a vascular wall component may be responsible for the observed leakage of macromolecules. This component does not appear to be the vascular smooth muscle itself. Two other elements of the vascular wall, the endothelium and the pericytes, have been shown to contain many of the same elements of the contractile machinery present in smooth muscle. Most recent studies have presumed that endothelial cell contraction is responsible for the formation of the interendothelial gaps through which the macromolecules move. However, endothelial contraction has been difficult to demonstrate experimentally. Alternatively, inasmuch as pericytic processes can end near endothelial junctions and there is an abundance of fibronectin between the pericytes and the endothelium, it may be a pericytic contraction that causes the interendothelial gap formation. — Miller, F. N.; Sims, D. E. Contractile elements in the regulation of macromolecular permeability. *Federation Proc.* 45: 84–88; 1986.

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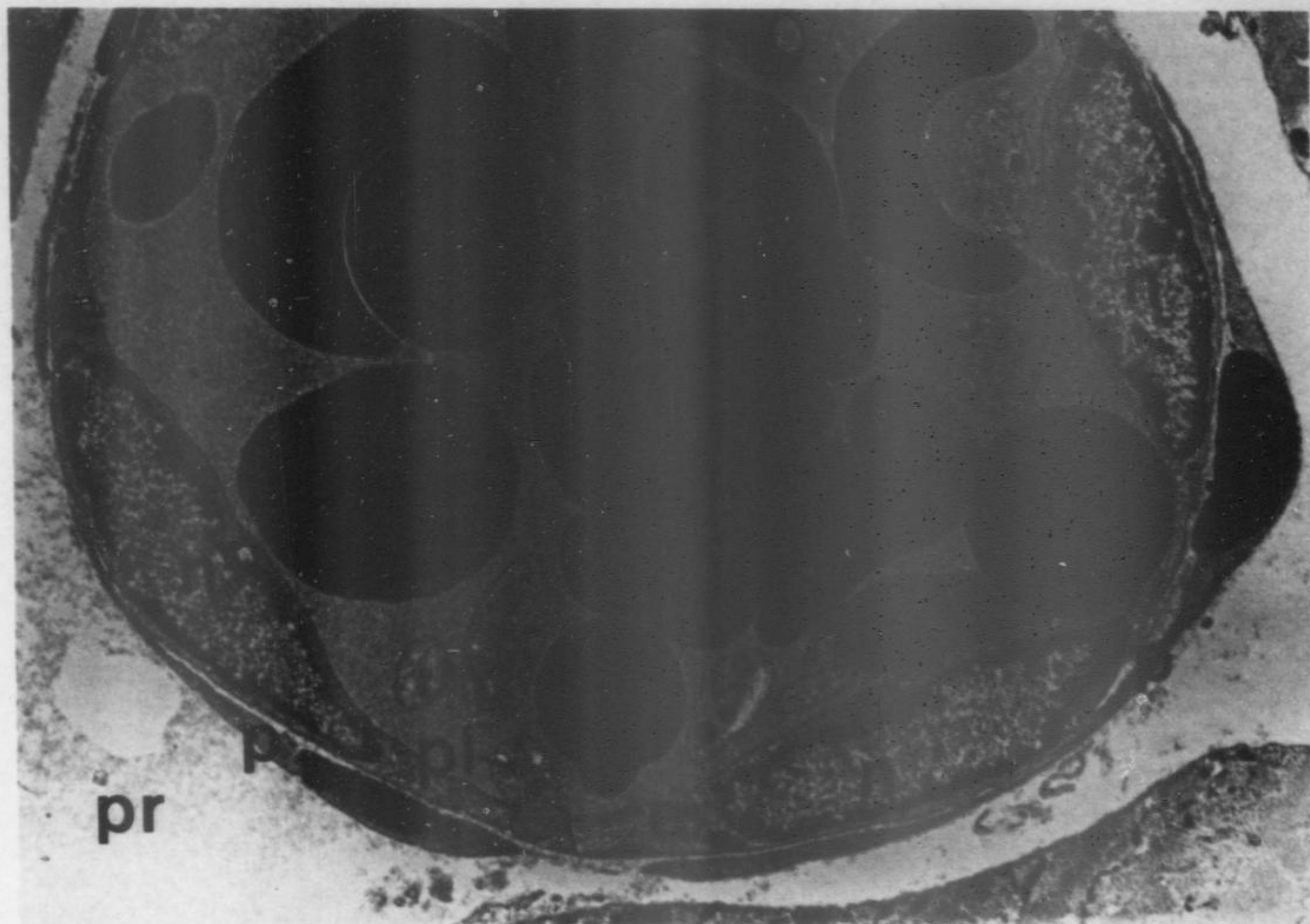


Figure 1. Venule (30 μm in diameter) from a rat cremaster muscle that had been treated with histamine (10^{-3} M). A platelet (pl) can be seen within an interendothelial gap. Red blood cells (r) have been trapped between pericytes (p) and endothelial cells (e). Proteinaceous material (pr) is in the interstitium. The nuclei (n) of endothelial cells have a wrinkled appearance, which suggests contraction of the cell.

phological studies have shown that gaps form between endothelial cells of venules when histamine is applied (10, 17). Thus, interendothelial gaps in venules are a very probable pathway for passage of the macromolecules. Inasmuch as Rippe and Grega (41) have shown that cooling inhibits macromolecular leakage produced by histamine in the rat hindquarters, gap formation and protein leakage are in all probability produced by an active process such as contraction of endothelium or pericytes. Demonstrating *in vivo* contraction of endothelium is a very difficult task, although there is evidence for contraction of cultured aortic endothelial cells (14). Mayhan and Joyner (34) have approached this problem by studying, *in vivo*, agents that are able to alter the contractile state of smooth muscle. They examined the effect of the calcium antagonist verapamil on the response to histamine in the hamster cheek pouch prepara-

tion. Either verapamil or the absence of calcium in the bathing solution inhibited the effect of histamine on macromolecular leakage. However, these types of studies demonstrate only that calcium is required for the effect, not that the endothelium is involved in the response.

Mechanisms of macromolecular leakage may also involve arachidonic metabolites because PGs can augment the responses to histamine and bradykinin (2, 48). This may be a direct interaction because histamine releases PGs from endothelial cells (1) and induces PGE_1 synthesis in rabbit ear preparations (31), whereas bradykinin increases calcium influx, resulting in activation of phospholipase A_2 (30). In addition, the cyclooxygenase inhibitors indomethacin and mefenamic acid have been shown to inhibit protein leakage induced by histamine (3). It is interesting that these drugs did not influence the vasodilatory effect of histamine on

the arterioles (3). The observation that arterioles are not affected by cyclooxygenase inhibitors supports the findings of Van de Voorde and Leusen (46) that endothelial-dependent relaxation in arteries is unaffected by indomethacin, but inhibited by eicosatetraynoic acid, quinacrine, and the H-1 antagonist mepyramine. These experiments indicate that the primary effect of histamine on arterioles is an endothelial-dependent, lipoxygenase-mediated relaxation.

In veins, endothelial-dependent relaxation of precontracted vessels has been difficult to demonstrate (15, 19), and *in vivo* venous dilation of small veins has not been observed even during mediator-induced protein leakage (3, 37). Thus, in venules the major effect of histamine may be an endothelium-dependent, cyclooxygenase-mediated event (3) that results in protein leakage but not vasodilation. However, other experiments demonstrate a very potent

effect of leukotrienes to produce arteriolar constriction followed by protein leakage (13, 22), which suggests that this is indeed a complicated story.

Common themes of many of these experiments include a role of calcium and an influence on contractile activity by compounds with different molecular structures. Thus, although smooth muscle itself is probably not a major factor in the induction of protein leakage by mediators of edema (5, 21, 37), it is likely that a contractile system similar to that found in smooth muscle is important (3, 25, 33, 34). The two primary effector candidates in the vascular wall other than vascular smooth muscle that could mediate macromolecular leakage are the endothelial cell and the pericyte (Fig. 1). A primary question has been, Do these cell types have contractile capability?

In the endothelium, immunofluorescent techniques (7) have shown that actin and myosin are present. Becker (6) has also described the presence of tropomyosin in various vascular endothelia. Drenckhahn (16) has reviewed studies that demonstrate that not only do the endothelial cells contain actin and myosin, but within the venular endothelium, actin and myosin filaments are closely associated with endothelial junctions. These filaments would be in an appropriate place to influence endothelial gap size. However, in a recent review Hammersen et al. (23) conclude that endothelial contractility has not been well demonstrated. There does not appear to be an anatomical explanation of the endothelium that would explain the heterogeneity of the protein leakage described by many investigators (17, 20, 37).

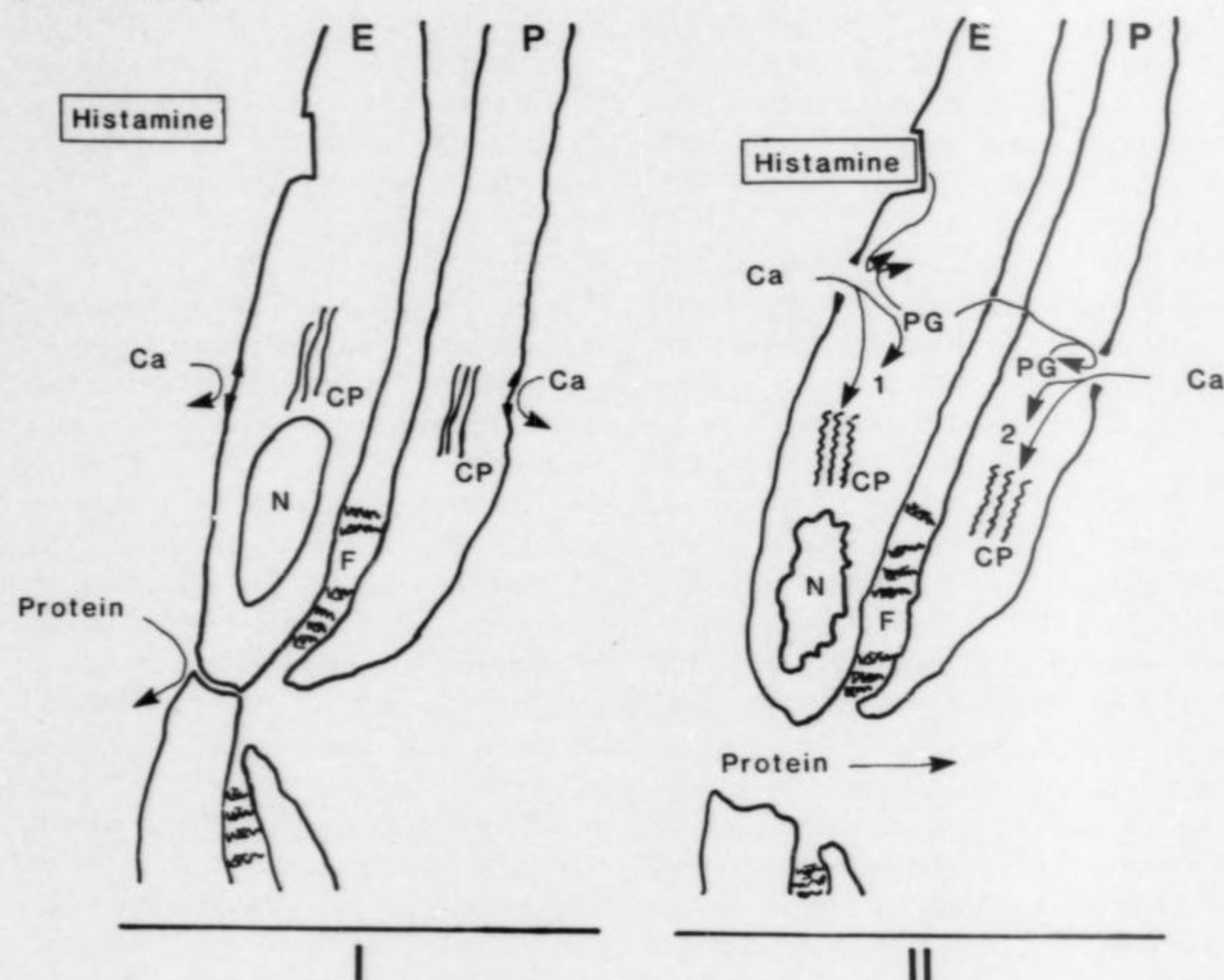
The other major vascular wall element that could be contractile and affect protein leakage is the pericyte. Pericytes encircle small arterioles, capillaries, and most venules, and are found in a wide variety of species and tissues (39, 40, 49). Studies with the scanning electron microscope (18, 35) have demonstrated that the pericyte is a long, irregular cell averaging 77 μm in length in rat skeletal muscle. It often has two or

more major processes and many cytoplasmic branches. However, pericytes do differ depending on the location (39, 40). Pericytes from rat skeletal muscle have processes that are larger and encompass twice the capillary circumference as those in the rat heart (44). Not only do pericytes differ from one tissue to another, but Fujiwara and Uehara (18) have shown that within a single vascular bed arterial, capillary, and venular pericytes have different morphologies. They are clustered and most numerous in the postcapillary venules (those that leak macromolecules in response to histamine and bradykinin) (18).

It has been recognized for many years that pericytes have a morphology suggestive of smooth muscle or a contractile cell, yet there is a paucity of direct evidence of their function and contractile nature. Tilton et al.

(45) demonstrated a selective wrinkling or buckling of the capillary endothelial cells directly under the pericyte processes in rat skeletal muscle preparations after applications of angiotensin, norepinephrine, and vasopressin. However, there have been no other recent demonstrations or suggestions of a contractile role for pericytes. Pericytes have been shown to contain mitochondria (8), smooth muscle-like actin (32), myosin (36), tropomyosin (27), and an enzyme related to smooth muscle function, GMP-dependent protein kinase (26). Furthermore, pericytes of small capillaries have predominantly a nonmuscle myosin whereas pericytes of postcapillary venules have a smooth muscle-like myosin (28). If pericytes function as the effector for mediators of edema, this could explain why macromolecular leakage has been observed only in

Figure 2. Model of the possible effects of histamine at an interendothelial junctional area. In panel I, the endothelial cell (E) with its nucleus (N) and contractile proteins (CP) and the pericyte (P) with its contractile proteins (CP) are quiescent because histamine has not attached to its receptor. Fibronectin (F) is found between the pericyte and endothelium and may serve to bind them together. Protein cannot move through the narrow gap and calcium (Ca) is restricted from moving into the cell. In panel II, histamine reacts with its receptor and induces calcium entry, which may result in contraction of the endothelium (pathway 1) and protein leakage. Histamine and calcium may also stimulate PG synthesis, which results in more calcium entry. PG, which moves across the cell membranes into the pericyte, could then induce calcium entry in the pericyte, which results in contraction (pathway 2) and protein leakage.



venules, and not in capillaries (17, 37, 38).

The anatomical location of pericytes and pericytic processes also portends an involvement in a heterogeneous response. For instance, pericytes of the gas exchange capillaries of the bovine lung (42) are frequently located near endothelial cell junctions. The proximity of the margins of pericytes to endothelial cell junctions suggests that whether or not pericytes are contractile, they may function in connection with endothelial cell junctions. Cultured pericytes produce collagen (11), and thus probably contribute to the basal lamina of microvessels. It is also known that the basal lamina between the pericytes and endothelial cells contains an abundance of fibronectin (12), an adhering substance that could connect the pericyte and endothelium. The potential role of fibronectin as a link between the endothelium and the pericyte could be crucial to understanding an endothelial cell-pericyte relationship. This interaction could be a structural bond of the intact fibronectin molecule from the

pericyte to the endothelium, or it could involve a regulatory role of certain fibronectin fragments that have been shown to have various biological actions (47), such as chemotaxis or opsonization.

It is evident that both endothelial cells and pericytes may be capable of contractile activity. The major hypothesis to explain macromolecular leakage has been that endothelial contraction results in the formation of interendothelial gaps through which the protein can easily move. This has proved very difficult to demonstrate. Alternatively, the literature is now equally supportive of a contractile role for the pericytes. These two hypotheses are diagrammed in Fig. 2.

Both of these hypotheses (endothelial or pericytic contraction) require a receptor type of interaction to initiate the process, and indeed histamine receptors have been demonstrated on the venular endothelial membrane (24). Thus, stimulation of the endothelium by histamine may result in the entry of calcium (30) and/or the release of cyclooxygenase products

(1, 31), which could lead to either endothelial or pericytic contractions. Fibronectin bonds from the pericyte to the endothelium (12) would allow the pericyte contraction to pull on the endothelium and produce the interendothelial gaps for protein leakage.

This type of cell-to-cell communication is well documented. For example, the effect of many agonists on vascular smooth muscle results from the release of substances from the endothelial cell and not from a direct effect of the agonist on the smooth muscle itself (15, 19). The heterogeneous distribution of pericytes and their processes could offer an anatomical explanation of why protein leakage occurs in discrete areas of some but not all venules after exposure to mediators of edema (17, 34, 37, 38). Although pericytic function has usually been discussed in terms of regulation of capillary blood flow (35) or phagocytosis (9), the proximity of the pericyte to both the endothelium and the interstitium places it in a unique position to also influence protein movement. E

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